Air shower

Part A. Electromagnetic shower (2.0 points).

A.0 (0.2pt)

$d_{\mathrm{e}} =$

A.1 (0.4pt)

$n_{\max} =$

A.2 (0.4pt)

$l_{\mathrm{e}} =$

A.3 (0.2pt)

$N_{\max} =$

A.4 (0.8pt)

$Q_{\mathrm{e}} =$

Part B. Hadronic shower (2.0 points).

B.1 (0.4pt)

$n_{\mathrm{hadr}}^{\max} =$

B.2 (1.0pt)

$n_{\mathrm{EM}}^{\max} =$

B.3 (0.6pt)

$\beta_{\mathrm{miss}} =$

Part C. Air shower (3.0 points).

C.1 (1.3pt)

$h_{\mathrm{stop}} =$

C.2 (0.5pt)

$\beta^{\mathrm{EM}} =$

C.3 (0.2pt)

$R_{\mathrm{stop}} =$

C.4 (0.2pt)

$S =$

C.5 (0.8pt)

$\Delta t =$

Part D. Data Analysis (3.0 points).

D.1 (1.0pt)

$\kappa =$

D.2 (0.5pt)

$E_0 =$

D.3 (1.5pt)

$\rho_{\mathrm{atm}} =$